

Chapter 17: Colorings of Graphs on Surfaces

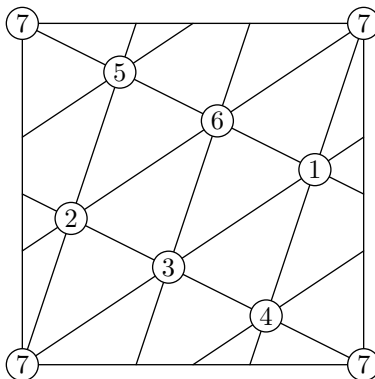
17.1 Genus and Orientable Surfaces

There are two types of surfaces: orientable and nonorientable. We consider here only the orientable type.

The plane and the sphere are equivalent for embedding purposes. In general one can take the sphere and add h handles. The resultant surface is called S_h and h is the **genus** of the surface. Equivalently the genus is the number of holes. The **torus** (or donut) has genus 1 and the pretzel has genus 2. The **genus** of a graph is the minimum genus of a surface needed to embed the graph. (Exists since worst-case one adds a handle for each edge.)

Now, it is important that there are two types of cycles on the surface. A **contractible** cycle can be continuously deformed or contracted to a single point. In the plane, all cycles are contractible, but on other surfaces they are not. If you slice through the torus on a noncontractible cycle, this will break the handle and you get a sphere; if you slice through the torus on a contractible cycle, you will get two pieces, a sphere and a torus.

For example, a torus can be obtained from a **rectangle** where the top and bottom are glued together, while the left and right are glued together. We say that the opposites sides are identified. Here is K_7 embedded on the torus, using this picture: the vertex at all four corners is the same vertex. In this example, cycle 136 is contractible, but cycle 123 is noncontractible.



There is an analogue of Euler's formula for a suitable definition of embedding. An embedding is a **2-cell** embedding if the boundary of each face is a contractible cycle.

THEOREM. *For a 2-cell embedding of a graph G on S_h , it holds that $V - E + F = 2 - 2h$ where V =vertices, E =edges, F =faces.*

PROOF. The proof is by induction on h . We already saw the result for $h = 0$.

So consider general S_h . Draw a noncontractible cycle C on S_h that does not go through any vertex of G . Since we have a 2-cell embedding, the cycle does not lie completely inside a face, and hence intersects some of the edges of G .

Now form a new graph G' by (a) adding vertices at every intersection between C and edges of G , and (b) adding edges embedded along C . The result is still a 2-cell embedding on the same surface.

Say there were k intersections between C and edges of G . Then G' has $V + k$ vertices, $E + 2k$ edges (k subdivisions and k new edges) and $F + k$ faces.

Now, cut S_h along the middle of C , thereby breaking the handle—called a capping operation. At the same time, duplicate each vertex and edge of C . The result is a graph G'' that has $V + 2k$ vertices, $E + 3k$ edges. Note that the cut exposes two new faces, bounded by the two copies of C . So the embedding has $F + k + 2$ faces, while it is on the surface with one less handle. It follows by the induction hypothesis that

$$(V + 2k) - (E + 3k) + (F + k + 2) = 2 - 2(h - 1).$$

And the result follows. $\square$

It follows that: $E \leq 3V + 6(h - 1)$.

17.2 Colorings

From Euler's formula, a graph on the torus has $E \leq 3V$ and so always has a vertex of degree at most 6. Hence

THEOREM. *The chromatic number of a toroidal graph is at most 7.*

This is best possible since K_7 has genus 1. Actually, any graph on the torus that requires 7 colors must contain K_7 as a subgraph. (Proof: induction/Brooks.)

In general, Euler's formula says that for any fixed surface, the average degree of a graph is $6 + o(1)$. From this there is an immediate upper bound on the chromatic number, by taking out a vertex of minimum degree and inducting. This bound turns out to be best possible. The hard part is showing that the largest clique that Euler's formula allows actually embeds on the surface.

There are also generalized coloring results for surfaces. For example, for the torus it is known that can partition the vertices of a toroidal graph into 5 sets such that each color component is either K_1 or K_2 . That is, each vertex has at most one neighbor of the same color. It is conjectured that the 5 can be replaced by 4. Another conjecture is that there always exists a triple of vertices such that when they are removed what remains is 4-colorable. In other words, yes seven colors are needed for the torus, but three of the colors need only be used once. (Note that this implies the 4-color theorem.)